Problem: Prove that the fraction $\frac{21n+4}{14n+3}$ is irreducible for every natural number n. (Source: 1959 IMO)

Solution: According to Bézout's identity, $21n+4$ and $14n+3$ are relatively prime if and only if there exist integers x and y such that $(21n+4)x+(14n+3)y=1$.

Let $x=-2$ and $y=3$.

$$\begin{aligned}(21n+4)x+(14n+3)y &= (21n+4)(-2)+(14n+3)(3)\\ &= -42n-8+42n+9\\ &= 1\end{aligned}$$

It follows, from Bézout's identity, that $21n+4$ and $14n+3$ are relatively prime.

Thus the fraction $\frac{21n+4}{14n+3}$ is irreducible for every natural number n.